8.8GHz Intel Core i9-13900K, Breaks World Record



Hardcore overclocker Elmor officially broke the CPU frequency world record with Intel's brand-new Core i9-13900K 24-core processor. The best CPU hit a staggering 8.812GHz using liquid nitrogen cooling, dethroning the 8-year reigning champion, the FX-8370, by 90MHz.

That's right; it took eight years for a new CPU architecture to dethrone AMD's FX series processors. Those chips are infamous for their mediocre CPU performance at launch; however, these chips scaled incredibly well under liquid nitrogen overclocking.

www.tomshardware.com/news/i9-13900k-world-record-overclock

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Great Ideas in Computer Architecture

(a.k.a. Machine Structures)



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Caches IV

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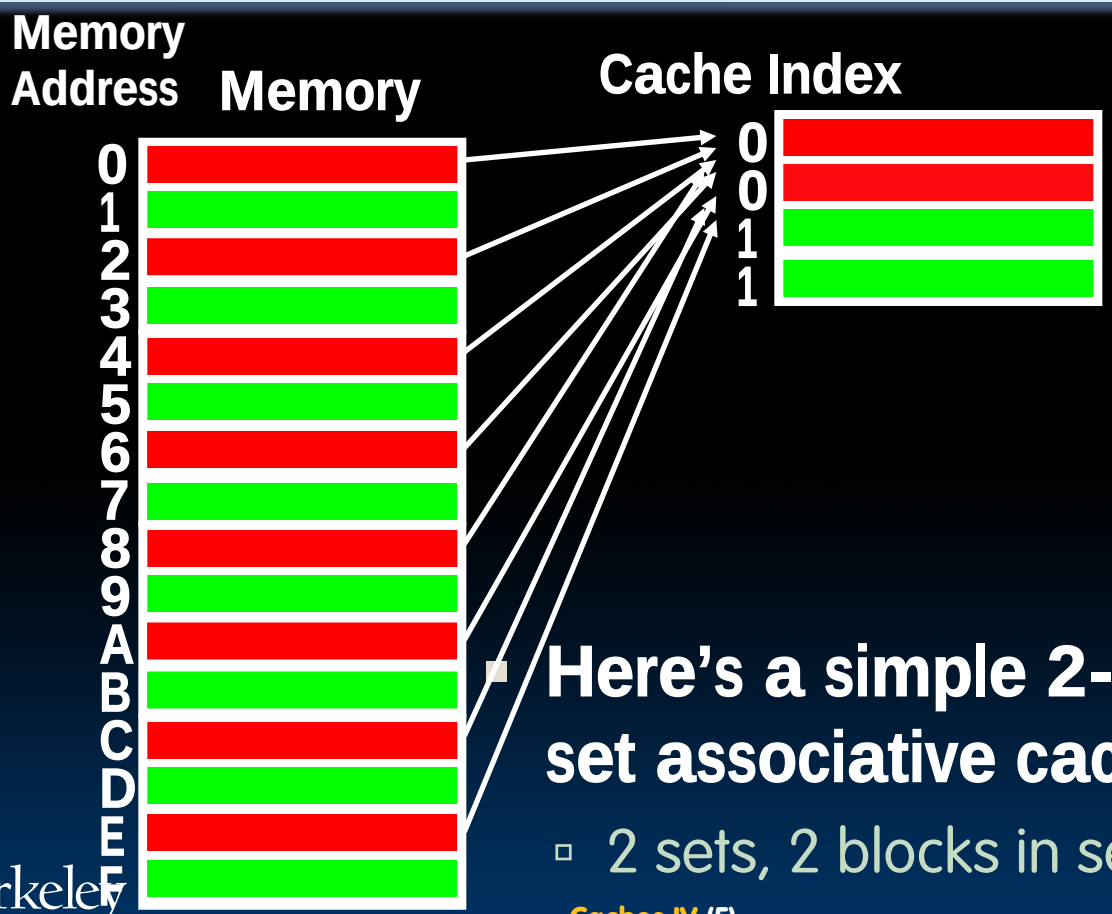
Set-Associative Caches

N-Way Set Associative Cache (1/3)

- **Memory address fields:**
  - Tag: same as before
  - Offset: same as before
  - Index: points us to the correct "row" (called a set in this case)
- **So what's the difference?**
  - each set contains multiple blocks
  - once we've found correct set, must compare with all tags in that set to find our data
  - Size of $ is # sets x N blocks/set x block size

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Associative Cache Example



- Here's a simple 2-way set associative cache.
  - 2 sets, 2 blocks in set

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N-Way Set Associative Cache (2/3)

- **Basic Idea**
  - cache is direct-mapped w/respect to sets
  - each set is fully associative with N blocks in it
- **Given memory address:**
  - Find correct set using Index value.
  - Compare Tag with all Tag values in that set.
  - If a match occurs, hit!, otherwise a miss.
  - Finally, use the offset field as usual to find the desired data within the block.

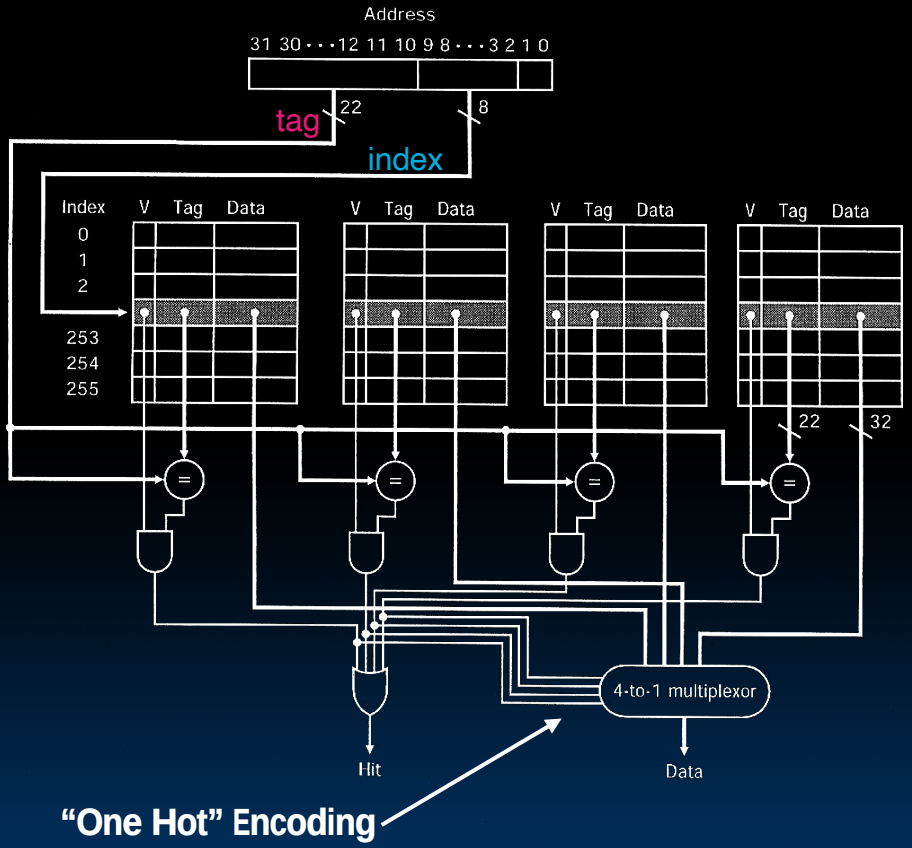
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N-Way Set Associative Cache (3/3)

- **What's so great about this?**
  - even a 2-way set assoc cache avoids a lot of conflict misses
  - hardware cost isn't that bad: only need N comparators
- **In fact, for a cache with M blocks,**
  - it's Direct-Mapped if it's 1-way set assoc
  - it's Fully Assoc if it's M-way set assoc
  - so these two are just special cases of the more general set associative design

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4-Way Set Associative Cache Circuit



"One Hot" Encoding

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Block Replacement with Example

Block Replacement Policy

- **Direct-Mapped Cache**
  - index completely specifies position which position a block can go in on a miss
- **N-Way Set Assoc**
  - index specifies a set, but block can occupy any position within the set on a miss
- **Fully Associative**
  - block can be written into any position
- **Question: if we have the choice, where should we write an incoming block?**
  - If there's a valid bit off, write new block into first invalid.
  - If all are valid, pick a replacement policy
    - rule for which block gets "cached out" on a miss.

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Block Replacement Policy

- **LRU (Least Recently Used)**
  - Idea: cache out block which has been accessed (read or write) least recently
  - Pro: temporal locality ➔ recent past use implies likely future use: in fact, this is a very effective policy
  - Con: with 2-way set assoc, easy to keep track (one LRU bit); with 4-way or greater, requires complicated hardware and much time to keep track of this
- **FIFO**
  - Idea: ignores accesses, just tracks initial order
- **Random**
  - If low temporal locality of workload, works ok

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Block Replacement Example

- Our same 2-way set associative cache with a four byte total capacity and one byte blocks. We perform the following byte accesses:

  0, 2, 0, 1, 4, 0, 2, 3, 5, 4

- How many hits and how many misses will there be for the LRU replacement policy?



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Block Replacement Example: LRU

0: miss, bring into set 0 (loc 0)

2: miss, bring into set 0 (loc 1)

0: hit

1: miss, bring into set 1 (loc 0)

4: miss, bring into set 0 (loc 1, replace 2)

Addresses 0, 2, 0, 1, 4, 0, ...

0: hit



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Cache Simulator!

www.ecs.umass.edu/ece/koren/architecture/Cache/frame1.htm

Block Replacement Simulator

Cache Size 4 # Sets 2

LRU
FIFO
RAND
Replacement Policy

Enter Query Sequence - Task A for Multi-tasking

in Decimal, or Hex

0 2 0 1 4 0

SHOW CACHE HELP

Set Repeat 2 cycles

Task B (when multi-tasking)

Cache Contents: LRU replacement policy;

Set#	4 Block, 2-way set-associative cache - tags shown in red	
0	0 - 00000000	4 - 00000010
1	1 - 00000000	-

COLOR KEY:	Compulsory Miss	Capacity Miss	Conflict Miss	Cache Hit	Unused

Cache Query Results:

Compulsory Misses :	3	Total Cache Queries :	6
Capacity Misses :	0	Total Misses :	4
Conflict Misses :	1	Miss Rate :	66.67 %
Cache Hits	2	Hit Rate :	33.33 %

Cache Query Sequence Trace

Cache Query Sequence Trace Address data replaced on miss shown in blue subscript					
0	2	0	1	4 -2	0

Addresses 0, 2, 0, 1, 4, 0, ...

Should be compulsory

	loc 0	loc 1
set 0	0	lru
set 1		
set 0	lru 0	2
set 1		
set 0	0	lru 2
set 1		
set 0	0	lru 2
set 1	1	lru
set 0	lru 0	4
set 1	1	lru
set 0	0	lru 4
set 1	1	lru

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Average Memory Access Time (AMAT)

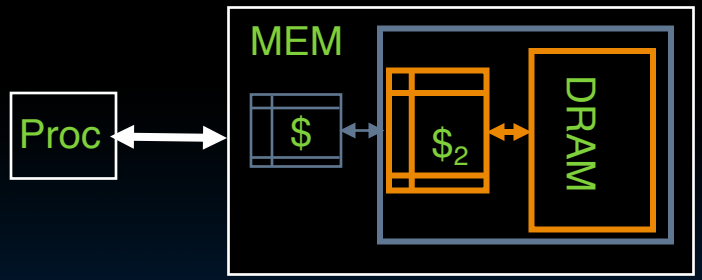
Big Idea

- **How to choose between associativity, block size, replacement & write policy?**
- **Design against a performance model**
  - Minimize: Average Memory Access Time
    = Hit Time
    + Miss Penalty x Miss Rate
  - influenced by technology & program behavior
- **Create the illusion of a memory that is large, cheap, and fast - on average**
- **How can we improve miss penalty?**

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Improving Miss Penalty

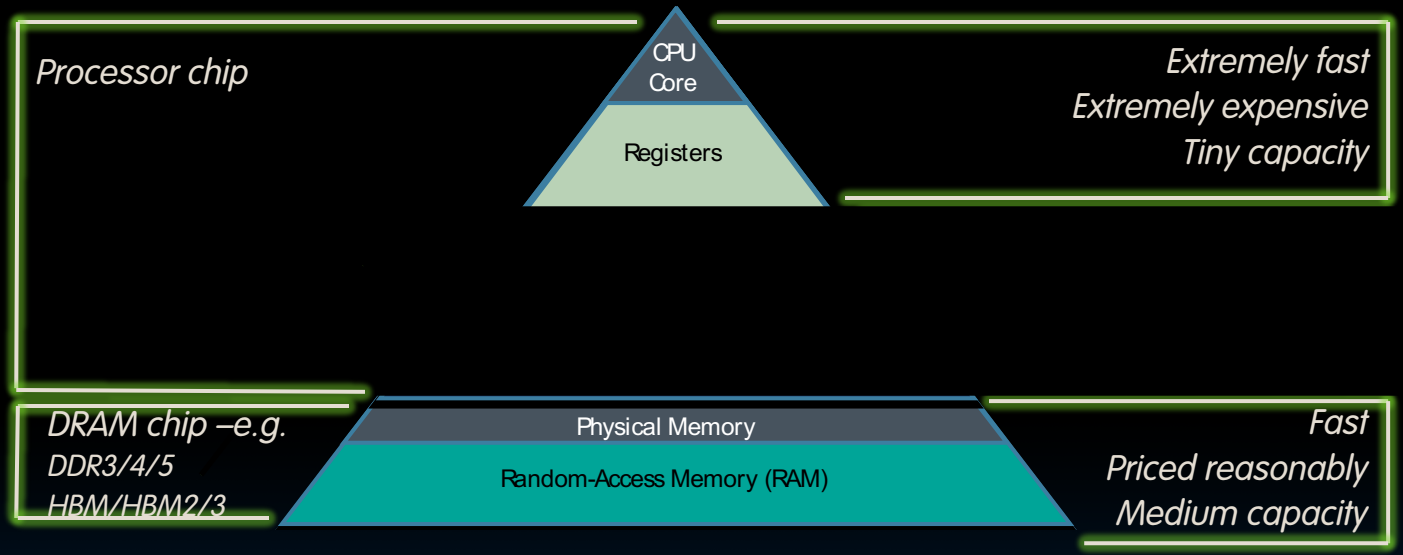
- When caches first became popular, Miss Penalty ~ 10 processor clock cycles
- Today 3 GHz Processor (1/3 ns per clock cycle) and 80 ns to go to DRAM
  ~200 processor clock cycles!



Solution: another cache between memory and the processor cache: Second Level (L2) Cache

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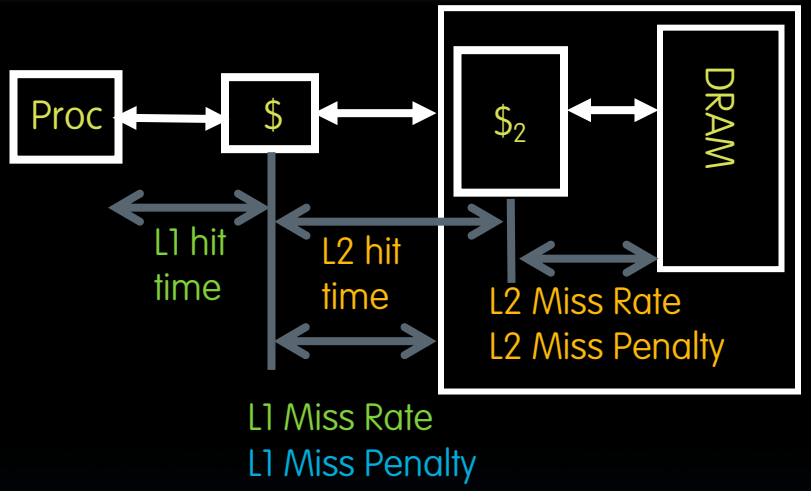
Great Idea #3: Principle of Locality / Memory Hierarchy



Let's see Cache configuration on Dan's computer…

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Analyzing Multi-level cache hierarchy



Avg Mem Access Time =
L1 Hit Time + L1 Miss Rate * L1 Miss Penalty

L1 Miss Penalty =
L2 Hit Time + L2 Miss Rate * L2 Miss Penalty

Avg Mem Access Time =
L1 Hit Time + L1 Miss Rate *
(L2 Hit Time + L2 Miss Rate * L2 Miss Penalty)

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Example

- **Assume**
  - Hit Time = 1 cycle
  - Miss rate = 5%
  - Miss penalty = 20 cycles
  - Calculate AMAT…
- **Avg mem access time**

  = 1 + 0.05 x 20

  = 1 + 1 cycles

  = 2 cycles

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Ways to reduce miss rate

- **Larger cache**
  - limited by cost and technology
  - hit time of first level cache < cycle time (bigger caches are slower)
- **More places in the cache to put each block of memory – associativity**
  - fully-associative
    - any block any line
  - N-way set associated
    - N places for each block
    - direct map: N=1

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Typical Scale

- **L1**
  - size: tens of KB
  - hit time: complete in one clock cycle
  - miss rates: 1-5%
- **L2:**
  - size: hundreds of KB
  - hit time: few clock cycles
  - miss rates: 10-20%
- **L2 miss rate is fraction of L1 misses that also miss in L2**
  - why so high?

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Example: with L2 cache

- **Assume**
  - L1 Hit Time = 1 cycle
  - L1 Miss rate = 5%
  - L2 Hit Time = 5 cycles
  - L2 Miss rate = 15% (% L1 misses that miss)
  - L2 Miss Penalty = 200 cycles
- **L1 miss penalty = 5 + 0.15 * 200 = 35**
- **Avg mem access time = 1 + 0.05 x 35**
  = 2.75 cycles

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Example: without L2 cache

- **Assume**
  - L1 Hit Time = 1 cycle
  - L1 Miss rate = 5%
  - L1 Miss Penalty = 200 cycles
- **Avg mem access time = 1 + 0.05 x 200**
  = 11 cycles

- **4x faster with L2 cache! (2.75 vs. 11)**

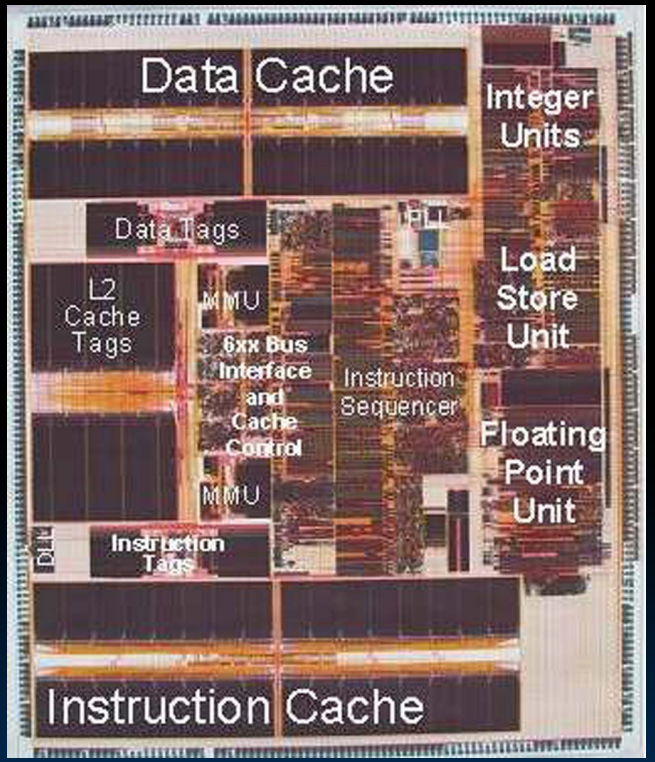


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Actual CPUs

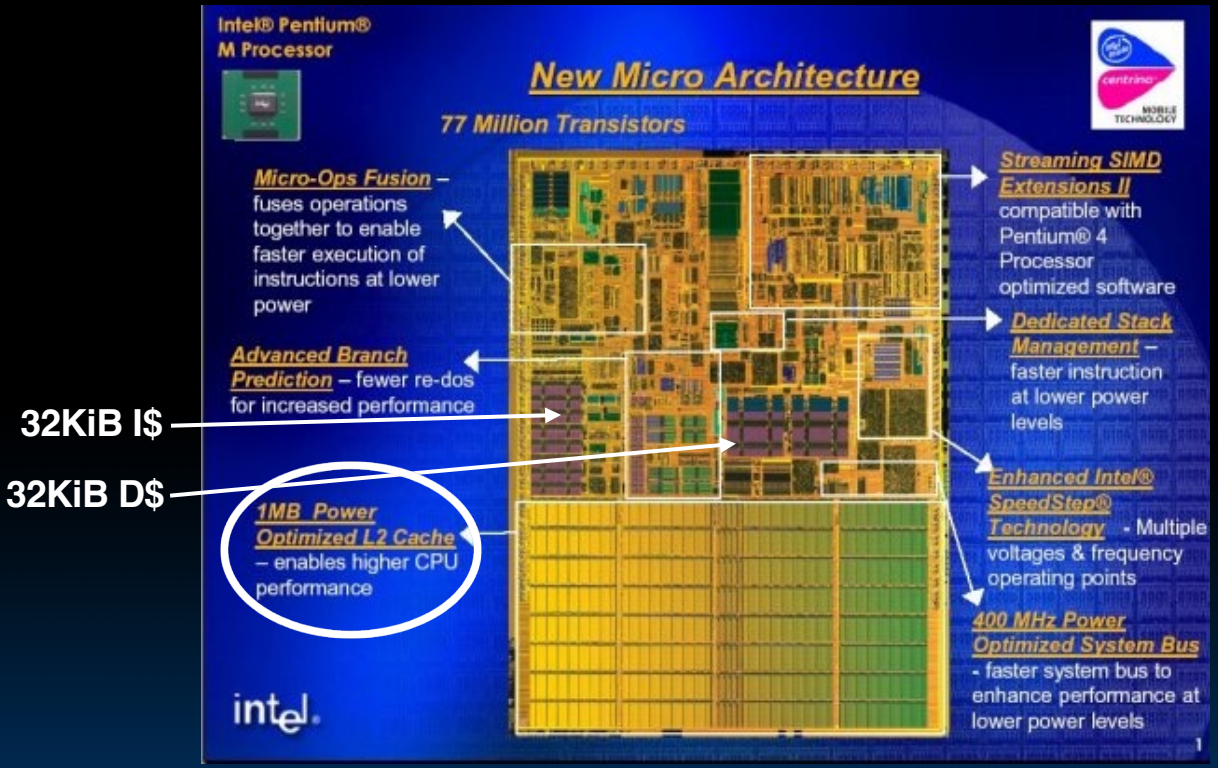
An Actual CPU – Early PowerPC

- **Cache**
  - 32 KiB Instructions & 32 KiB Data L1 caches
  - External L2 Cache interface with integrated controller and cache tags, supports up to 1 MiB external L2 cache
  - Dual Memory Management Units (MMU) with Translation Lookaside Buffers (TLB)
- **Pipelining**
  - Superscalar (3 inst/cycle)
  - 6 execution units (2 integer and 1 double precision IEEE floating point)



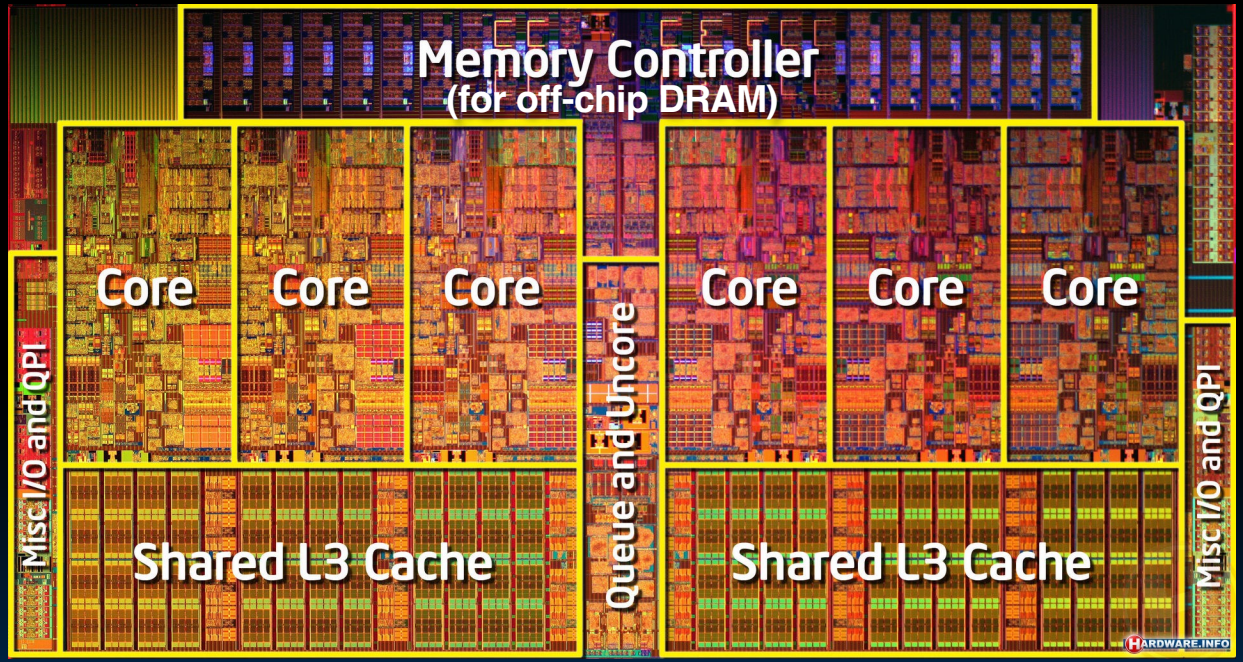
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An Actual CPU – Pentium M



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An Actual CPU – Intel core i7



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Text **DDG** to **22333** once to join

L27 A 2-way set-associative cache cannot be outperformed by a direct-mapped cache.

The worst case for N-way set-associative cache is a repeated pattern of N addresses that map to that set.

On recent chips L4 cache is on chip.



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How can DM beat 2-way?

- Sure, consider the caches from the previous slides (2-way set associative cache with a four-byte total capacity and one-byte blocks) with the following workload:

  0, 2, 0, 4, 2

2-way: 0m, 2m, 0h, 4m, 2m;

DM: 0m, 2m, 0h, 4m, 2h

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And in Conclusion…

- **We've discussed memory caching in detail. Caching in general shows up over and over in computer systems**
  - Filesystem cache, Web page cache, Game databases / tablebases, Software memoization, Others?
- **Big idea: if something is expensive but we want to do it repeatedly, do it once and cache the result.**
- **Cache design choices:**
  - Size of cache: speed v. capacity
  - Block size (i.e., cache aspect ratio)
  - Write Policy (Write through v. write back
  - Associativity choice of N (direct-mapped v. set v. fully associative)
  - Block replacement policy
  - 2nd level cache?
  - 3rd level cache?
- **Use performance model to pick between choices, depending on programs, technology, budget, ...**

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